Group Project Guidelines

1. Task

In this group project, your team is required to design and implement a multi-agent system to accomplish a practical task. You may choose your own application scenario. Possible directions include, but are not limited to:

- Software development and coding
- Information retrieval and research
- Question answering
- Planning and decision-making
- Data analysis
- Content generation
- Task automation
- Game or simulation environments
- Other applications where collaboration among multiple agents is beneficial

The project should demonstrate a clear understanding of LLM-based agents and multi-agent systems. Instead of simply invoking multiple LLMs independently, your system should include a meaningful mechanism for agent specialization, communication, coordination, or collaboration.

Each agent should have a clearly defined role or capability, and the overall system should demonstrate how multiple agents work together to accomplish the selected task. Possible designs may include, but are not limited to:

- Role-based agent collaboration
- Planner – executor architectures
- Debate or discussion among agents
- Hierarchical agent systems
- Supervisor – worker architectures
- Agent routing and task delegation
- Shared memory or shared workspace
- Multi-agent reflection and verification
- Tool-using agents

The expected outcome of the project is a working multi-agent system that can be demonstrated on representative task examples.

Your team should clearly explain why a multi-agent approach is appropriate for the selected task, how the agents interact, and what benefits the proposed design provides.

2. Requirements

Each group should complete the design, implementation, and demonstration of the proposed multi-agent system.

System Requirements

Your project should:

1) Address a clearly defined task or application scenario.

2) Implement a multi-agent architecture with multiple agents playing meaningful and distinguishable roles.

3) Clearly define:
- The role and responsibility of each agent;
- The information or tools available to each agent;
- How tasks are assigned or decomposed;
- How agents communicate or exchange information;
- How the final output or decision is produced.

4) Provide a working implementation that can be demonstrated using representative examples.

3. Submission Materials

Each group is required to submit the following materials:

- System source code, together with necessary instructions for running the system;
- Demo video, demonstrating the overall workflow and representative examples. The video should be no longer than 10 minutes;
- Presentation slides.
- Individual Contribution Statement, briefly describing the module(s) or component(s) each member is responsible for and how their work interacts with or contributes to other members' modules.

The presentation slides should briefly cover:

1) Problem & Motivation

What task does the project address, and why is a multi-agent approach suitable?

2) Multi-Agent System Design

What agents are included, what are their roles, and how do they collaborate?

3) Implementation

Briefly introduce the main models, tools, frameworks, or techniques used.

4) System Demonstration & Results

Show representative examples of how the system works and briefly discuss its performance or limitations.

Please refer to the detailed submission instructions on Canvas and prepare your submission before the deadline:

Submission Deadline: November 23, 2026, 11:59 PM

Please submit your project materials through the corresponding Assignment section on Canvas. Only one member from each group needs to upload the submission on behalf of the whole group.

It is recommended that all submission materials be packaged together into a single compressed file (e.g., .zip) before uploading.

Please name the submission file using your group ID, for example:

COMP5584+GroupProject+Group5.zip

4. Grading Rubric

The final grade will take into account both the overall quality of the group project and each member's individual contribution. Individual grades may be adjusted based on the contribution statement and demonstrated involvement in the project. Members who

make no substantive contribution will receive no marks.

Criterion Weight Description:

1) Problem Formulation & Motivation (15%): Clarity and significance of the selected task; appropriateness and justification of using a multi-agent approach.

2) Multi-Agent System Design (30%): Clear agent roles, task decomposition, communication/coordination mechanisms, and overall architecture. The design should demonstrate meaningful multi-agent collaboration.

3) Technical Implementation & Functionality (25%): Correctness and completeness of the implementation, appropriate use of LLM/agent techniques, system functionality, and quality of the demonstrated system.

4) Analysis & Innovation (15%): Technical insight, creativity, analysis of system behavior, limitations, failure cases, and design rationale.

5) Presentation & Submission Quality (15%): Clarity and organization of the presentation, quality of the demo video and slides, and effectiveness in communicating the project.

A technically complicated system does not necessarily receive a higher score. Greater emphasis will be placed on whether the multi-agent design is well motivated, clearly implemented, and genuinely contributes to solving the selected task.

5. Consultation

You may send questions about the group project to the teaching assistant, Qi Peng (qi0125.peng@connect.polyu.hk).

To avoid your email being overlooked, it is strongly recommended that you begin your email subject line with: “COMP5584+GroupProject+Group[ID]”
(e.g., “COMP5584+GroupProject+Group5”).